
FINITE-SAMPLE SAFE ADAPTIVE EARLY STOPPING FOR CORRELATED MAJORITY-VOTE ROLLOUTS

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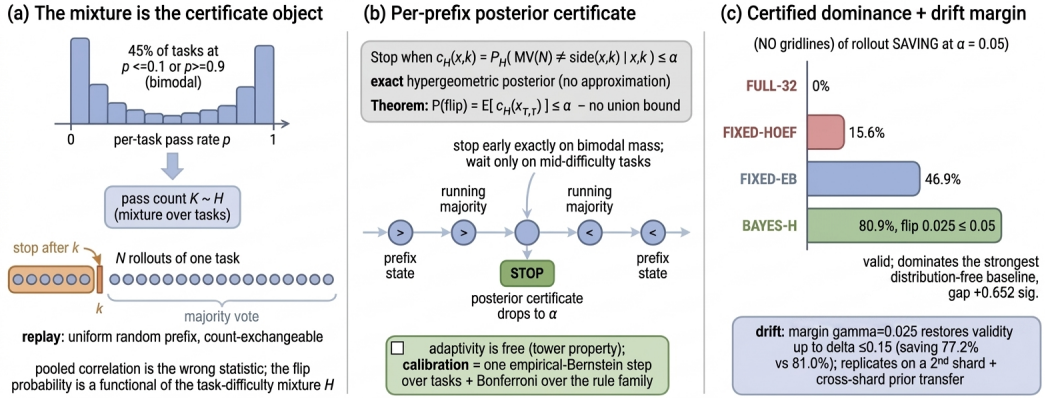
ABSTRACT

Repeatedly sampling a model and taking a majority vote (self-consistency) is a standard way to spend inference compute, but the vote rarely needs all N samples. Existing adaptive stoppers either assume independent samples or use heuristic window rules with no error control. We give a *mixture-posterior stopper* (BAYES-H) whose per-problem conditional flip certificate is exact: given a calibration prior H over per-task pass counts, stop as soon as the posterior probability that the full- N majority will disagree with the current prefix majority drops below α . By the tower property the population flip rate of any such rule is bounded by α *without* a union bound over stopping times (adaptive is free), and the unknown-mixture layer adds only an empirical-Bernstein calibration step over problems, with Bonferroni control of the whole searched rule family. On 11,607 math problems with $N = 32$ rollouts each (OpenMathReasoning), selected and evaluated under a strict FIT/CAL/TEST problem split with a single held-out readout, BAYES-H saves 81% of rollouts at realized flip rate $0.025 \leq \alpha = 0.05$ and 73% at $\alpha = 0.02$, dominating the strongest distribution-free baseline (Hoeffding-fixed- k : 16% at $\alpha = 0.05$) with a statistically significant paired gap, and the strongest mixture-aware fixed budget (empirical-Bernstein fixed- k : 47%). An ESC-style window heuristic that saves 86% violates its target ($0.058 > 0.05$). Under pre-fixed ordered-drift stress mechanisms (front-loading, linear trend, adversarial block swap), BAYES-H degrades the most gracefully of all stoppers, and a simple certificate margin restores validity up to a quantified drift budget. All conclusions replicate on a disjoint second shard, and survive transferring the calibration prior across shards (total-variation 0.042) with ≤ 0.4 points of movement. Two further model-carrier pairs bracket the mechanism: on OpenR1-Math-220k (DeepSeek-R1, $M = 2$ per problem) the certificate attains its ordered frontier exactly when the mixture structure allows, and never over-stops; on RLVE (Qwen3-4B, $N = 8$ per problem, verifiable environments) it lands at the intermediate point the mechanism predicts — 51–58% savings at $\alpha \in \{0.10, 0.05\}$, 45–48% at $\alpha \in \{0.02, 0.01\}$, certified at *every* level down to $\alpha = 0.01$ where *no* distribution-free budget is certifiable.

1 INTRODUCTION

Self-consistency (Wang et al., 2023) replaces one sampled answer by a majority vote over N samples. Adaptive variants spend fewer samples on easy inputs: ASC (Aggarwal et al., 2023) chooses a per-problem sample count from a confidence proxy, ESC (Li et al., 2024) stops when a sliding window of samples agrees, and MARS (Chen et al., 2026) allocates a budget across problems of-fline. Two real effects are uncontrolled in this literature: (i) rollouts of the same task are *correlated* (shared prompt, shared sampling regime), so independence-based certificates are overconfident; and (ii) rollouts are generated *in sequence*, so distribution drift along the generation order can break any certificate built on exchangeability.

We make two structural observations that reframe the problem. First, on real carriers the dominant deviation from i.i.d. is not latent within-task correlation but *across-task heterogeneity*: per-task pass rates are strongly bimodal (Section 8), and the flip probability a stopper must control is a functional



Treat across-task heterogeneity as the certificate object (a): stop when the exact posterior flip probability drops to α (b) — adaptivity is free and the result saves 81% of rollouts where the strongest distribution-free baseline saves 16% (c).

Figure 1: **Overview.** (a) Treat across-task heterogeneity as the certificate object: per-task pass rates are strongly bimodal, so the flip probability is a functional of the difficulty mixture H , not of a pooled correlation. (b) BAYES-H stops when the exact posterior flip probability $c_H(x, k)$ drops to α ; by the tower property adaptivity is free (no union bound), and calibration is one empirical-Bernstein step over tasks. (c) At $\alpha = 0.05$ BAYES-H saves 80.9% of rollouts (realized flip 0.025), dominating the strongest distribution-free baseline (FIXED-HOEF, 15.6%) and the strongest mixture-aware fixed budget (FIXED-EB, 46.9%); a certificate margin restores validity under ordered drift, and all results replicate on a disjoint second shard. All numbers are verified artifacts (REPRO_README.md).

of the *mixture* of per-task difficulty, not of a pooled correlation coefficient. Pooled second-moment summaries can be simultaneously over-conservative and uncalibrated (Section 8). Second, once the target is the mixture, the correct object is a *per-prefix conditional certificate*: the posterior probability, given the observed prefix, that the full- N majority disagrees. This object is exactly computable (hypergeometric posterior), its fixed-budget expectation $\mathbb{E}_H[f_K(k)]$ is monotone in the prefix length (Lemma 1), and — because the tower property collapses the stopping-time dependence — it certifies any adaptive rule at level α with no union bound (Theorem 1).

Contributions.

1. **Framework.** A count-exchangeable replay model in which both within-task correlation and across-task heterogeneity are folded into a mixture H over per-task pass counts; the certificate target is the flip probability under H , unifying the two effects (Section 3).
2. **Method.** BAYES-H: stop when the posterior conditional flip probability $c_H(x, k) \leq \alpha$. Exact certificates (Lemma 1, Theorem 1); adaptivity is free — no union bound, no e-process tuning (Section 4).
3. **Legitimate selection.** A FIT/CAL/TEST problem-level protocol in which the rule family (budget $\times$ level for fixed- k ; level for BAYES-H) is selected on CAL with simultaneous empirical-Bernstein bounds (Maurer & Pontil, 2009) plus Bonferroni, and TEST is read once (Section 5).
4. **Empirical gains.** On 11,607 OpenMathReasoning problems, 80.9% rollout saving at realized flip 0.025 ($\alpha = 0.05$), vs 15.6% for the strongest distribution-free baseline at the same level; ESC-style window stopping is invalid ($0.058 > 0.05$). Gains are paired-CI significant (Section 6). Two further carriers with different models (DeepSeek-R1 at $M = 2$; Qwen3-4B at $N = 8$ on verifiable environments) show the certificate attains its ordered frontier (stop when valid, never when not) and interpolates with the resolvable mixture structure, i.e. the method is carrier-aware rather than a generic stop-early trick (Section 6).
5. **Honest boundaries.** The public carrier gives per-task pass *counts*, not ordered rollouts; it exactly supports random-prefix (count-exchangeable) certificates but cannot validate ordered on-line deployment. We therefore bound the failure domain under three pre-fixed ordered-drift mechanisms and quantify a certificate-margin repair (Section 7).

2 RELATED WORK

Adaptive sampling for voting. ASC (Aggarwal et al., 2023) stops sampling when a lightweight model’s answer distribution is confident; ESC (Li et al., 2024) stops when a window of consecutive samples agrees. Both lack finite-sample control of the disagreement-with-full-vote event; we show the ESC rule exceeds its nominal level on heterogeneous mixtures (Table 1). MARS (Chen et al., 2026) allocates sampling budget across problems offline with a learned router; our setting is online per-task stopping with certificates. Best-of- n and budget-forcing methods (Snell et al., 2025) tune compute without stopping certificates.

Conformal and anytime inference. Conformal prediction (Vovk et al., 2005) gives marginal coverage under exchangeability; confidence sequences (Howard et al., 2021) give anytime-valid bounds for i.i.d. means. Our certificate is neither: it is an *exact posterior flip probability* under a calibrated discrete prior, valid conditionally per prefix (hence adaptivity is free), with the mixture estimated from calibration problems under an empirical-Bernstein bound (Maurer & Pontil, 2009). The Bonferroni control of a small pre-declared rule family is deliberately conservative and transparent.

Dependent sampling. Design-effect corrections (Kish, 1965) shrink effective sample size by an intraclass correlation; we show pooled correlation is the wrong functional here (Section 8). Beta-binomial and correlated-Bernoulli models (Skellam, 1948) capture overdispersion but target estimation, not stopping certificates for the flip event.

3 MODEL: COUNT-EXCHANGEABLE REPLAY

A task’s N rollout outcomes are summarized by the pass count $K \in \{0, \dots, N\}$. Across tasks, $K \sim H$, a mixture capturing both task heterogeneity and any within-task correlation: the probability a fresh rollout passes, marginally, is $\mathbb{E}_H[K]/N$, and the joint law of the N outcomes is exchangeable *within* a task given its realized count. Deployment model (*replay*): the stopper observes a uniform random prefix drawn without replacement from the task’s N outcomes — i.e., the first k draws are exchangeable with the remaining $N - k$. Section 7 stress-tests departures from this model (ordered drift); our validity claims are stated relative to it, and the carrier limitation (counts only, no order) is made explicit in Section 9.

Let $MV(k) = \mathbf{1}\{x_k > k/2\}$ with ties assigned to the negative side, and $MV(N)$ the full-vote decision ($K > N/2$). The *flip event* of a stopping rule τ is $\{MV(\tau) \neq MV(N)\}$.

Remark 1 (Why the mixture, not the correlation). On τ^2 -bench carriers (6 public task suites, 4 rollouts/task) we measured pooled within-task reward correlation $\hat{\rho} = 0.47\text{--}0.59$, but a task-demeaned (fixed-effect) estimate gives $\hat{\rho}_{FE} \approx -1/3 \approx -1/(N - 1)$, the demeaning artifact floor: essentially *all* observed correlation is across-task heterogeneity, not latent within-task dependence. On OpenMathReasoning the per-task pass-rate variance exceeds the binomial baseline by $34\times$. The flip probability is driven by the mass of tasks with $K \approx N/2$ — a mixture functional — so pooled second-moment discounts are the wrong certificate statistic.

4 METHOD: THE MIXTURE-POSTERIOR STOPPER

Exact per-task flip probability. Given count K , the prefix count x_k is hypergeometric $HG(N, K, k)$, hence

$$f_K(k) := \mathbb{P}(MV(k) \neq MV(N) \mid K) = \sum_x HG(N, K, k, x) |\text{side}(x, k) - \text{side}(K, N)|. \quad (1)$$

Lemma 1 (Exactness and monotonicity). (1) *is an identity (no approximation). For every fixed K , $f_K(k)$ is nonincreasing in odd k ; hence for any mixture H , $\mathbb{E}_H[f_K(k)]$ is nonincreasing and $k^*(\alpha) = \min\{k : \mathbb{E}_H[f_K(k)] \leq \alpha\}$ is well defined and binary-searchable.*

Proof. The identity is a definition expansion. Monotonicity follows from a coupling of the prefixes at k and $k+2$ on a common ordered draw: the difference $f_K(k) - f_K(k+2)$ reduces to two boundary atoms whose closed-form ratio is $(K - t)/(N - K - t)$, nonnegative for $K > N/2$, with the complementary case $K \leq N/2$ handled by pass/fail reflection; the full proof is in Appendix A.

Independently, monotonicity is verified by exhaustive enumeration over all 33×15 (K, k) cells at $N = 32$ (zero violations, tolerance 10^{-12}). $\square$

Conditional certificate. Given prior H and an observed prefix (x, k) , define the posterior flip probability

$$c_H(x, k) := \mathbb{P}_H(\text{side}(K, N) \neq \text{side}(x, k) \mid x, k), \quad \text{posterior} \propto H[K] \cdot \text{HG}(N, K, k, x). \quad (2)$$

Theorem 1 (Adaptive is free). *Let τ be any stopping time measurable w.r.t. the prefix filtration that stops only at states with $c_H(x_\tau, \tau) \leq \alpha$ (forced stop at $k = N$ contributes its own conditional value). Then $\mathbb{P}(\text{MV}(\tau) \neq \text{MV}(N)) = \mathbb{E}[c_H(x_\tau, \tau)] \leq \alpha$, with equality in the identity and no union bound over stopping times.*

Proof. $\mathbb{P}(\text{flip}) = \mathbb{E}[\mathbb{P}(\text{flip} \mid \text{prefix path})]$. For any path stopped at (x, k) , $\mathbb{P}(\text{MV}(N) \neq \text{side}(x, k) \mid \text{path}) = c_H(x, k)$ since the posterior is exactly the conditional law of K and $\text{MV}(N)$ is a function of K . The rule bounds the integrand pointwise by α (the forced-stop boundary contributes $\mathbb{E}[c_H \mathbf{1}\{\tau = N\}]$, absorbed into the population bound of Theorem 2); taking expectations gives the claim. $\square$

BAYES-H is the rule $\tau = \min\{k \geq 3 : c_H(x_k, k) \leq \alpha\}$ with $H = \hat{H}_m$, the calibration empirical mixture. Its per-task exact flip probability and expected stopping time under any true count K are computable by a linear-time DP over prefix states (k, x) (Appendix B), enabling exact calibration and evaluation without Monte Carlo.

Theorem 2 (Population layer). *Let $g(K; \alpha, \hat{H}_m)$ be the exact per-task flip probability of BAYES-H under count K , and let $\bar{g}_{\text{CAL}}, s_{\text{CAL}}^2$ be its mean and variance over m calibration problems. With probability $1 - \delta$ over the calibration sample,*

$$\mathbb{P}_{\text{replay}, H}(\text{flip}) \leq \bar{g}_{\text{CAL}} + \sqrt{\frac{2s_{\text{CAL}}^2 \log(4/\delta)}{m}} + \frac{7 \log(4/\delta)}{3(m-1)},$$

by the empirical-Bernstein inequality (Maurer & Pontil, 2009) applied to the bounded per-task flips $g \in [0, 1]$, with δ split by Bonferroni over the pre-declared selection family ($J = 64$ candidates; Section 5).

Remark 2 (No Jensen gap). The population flip rate under replay is a linear functional of H : $\mathbb{P}(\text{flip}) = \sum_K H[K] g(K)$. Hence estimation error in $\hat{H}_m$ enters only through the empirical mean of a bounded per-task quantity — there is no Jensen correction, and empirical Bernstein is essentially tight at $m = 4000$.

Proposition 1 (Exact TV-robustness certificate). *Fix the rule (certificate tables frozen from $\hat{H}$) and let $g(K)$ be its exact per-task flip probability. Over the total-variation ball $B_R = \{q : q \geq 0, \sum_K q_K = 1, \|q - \hat{H}\|_1 \leq 2R\}$,*

$$V(R) := \max_{q \in B_R} \sum_K q_K g(K)$$

is a linear program whose optimum is a two-sided greedy: add mass to atoms in descending g (capped by $q_K \leq 1$) and drain mass from atoms in ascending g (capped by $\hat{H}_K$); $V(R)$ is hence computable in closed form in $O(N \log N)$, and it strictly refines the 1-Lipschitz bound $\text{flip}(\hat{H}) + R$ whenever the g -profile is non-constant. The critical radius $\tau^(\alpha) = \max\{R : V(R) \leq \alpha\}$ quantifies how much prior shift the frozen certificate provably tolerates at level α .*

On the OMR carrier the critical radii are $\tau^*(0.10/0.05/0.02/0.01) = 0.144/0.078/0.074/0.052$ (shard 0) and $0.157/0.082/0.076/0.024$ (shard 1) (tv_robustness_r481_result.json; the closed form matches a direct simplex LP to 10^{-6} on a $6 \times 4 \times 2$ grid of radii, levels and shards). Three consequences: (i) at the observed cross-shard mismatch $R = 0.042$, the worst-case flip over the whole ball is $0.039 \leq \alpha = 0.05$ and $0.016 \leq 0.02$, so the cross-shard transfer of Section 6 is certified *post hoc* on both shards — but on shard 1, $\tau^*(0.01) = 0.024 < 0.042$, i.e. TV alone does not certify the transfer at $\alpha = 0.01$, which is exactly the level the transfer experiment does not report; (ii) a *prior-shift validity band* follows directly: certifying the frozen tables at the tightened level $\alpha - R$ keeps the worst-case flip over B_R below α whenever the re-certified rule’s $V(R) \leq \alpha$; at $R = 0.05$, $\alpha = 0.10$ re-certified at 0.05 has worst-case flip $0.042 \leq 0.10$ and still saves 80.3% of rollouts (vs 83.6% without the band) — a 3.3-point price for validity uniform over a 5-point TV neighbourhood;

(iii) at every level a 36–47% mass of atoms has $g(K) = 0$ exactly (the certificate stops only on states the prefix already decides), which is why $V(R)$ grows far slower than the Lipschitz rate $R \cdot \max_K g(K)$. *Cross-carrier*. On the RLVE carrier ($N=8$; same frozen FIT prior as Section 6, no TEST re-read) the same closed form gives $\tau^*(0.10/0.05/0.02/0.01) = 0.199/0.551/0.213/0.449$ (tv_robustness_rlve_r482_result.json; LP cross-check matches to 10^{-6}). The radii are larger than on OMR at every level even though the RLVE prior is no more heterogeneous (het-excess variance 0.0999 vs 0.129): the RLVE prior concentrates 74–87% of its mass on zero- g atoms (vs 36–47% on OMR), and with only $N=8$ atoms the worst-case adversary has few high- g directions in which to move mass. Across carriers τ^* co-varies with the *alignment* of the prior with the zero- g set and is uncorrelated with heterogeneity (het-excess vs τ^* : Spearman 0.01; het-excess vs alignment: 0.05 — the two candidate drivers are not confounded). The alignment- τ^* co-variation is a *between-carrier* law: pooled Spearman 0.75 over the 16 carrier $\times$ level cells, 1.0 over the four carrier means, 0.63 with the two whole-simplex-plateau cells removed; *within* OMR it is negative ($-0.95/-0.89$ per shard — tightening the level moves the zero- g set only at the most-aggressive end, where mass is added on the lowest- g atoms), positive within RLVE (0.32) and OpenR1 (1.0) (decomposition audit: spearman_decomposition_r486_result.json; Appendix D). (Two disclosures: τ^* bisection uses $R \in [0, 1]$ here ($0.551 > 0.5$ exceeds the $[0, 0.5]$ range sufficient for OMR); τ^* is non-monotone in α — $\tau^*(0.05) > \tau^*(0.10)$ — because tightening α zeroes $K=6$ ($g : 0.107 \rightarrow 0$) and shrinks $K=5$ ($0.339 \rightarrow 0.071$), flattening the g -profile’s top faster than the budget tightens; internal drops exist as well (the curve has 7 rule-change breakpoints on the plotted range, Figure 2), so τ^* should not be read as globally monotone in α .) *Three-atom edge case*. On OpenR1 ($M=2$; same frozen FIT prior, no TEST re-read) the support is three atoms $p \in \{0, \frac{1}{2}, 1\}$ and $g(0) = g(1) = 0$ at every level (a decided prefix cannot be overturned by one more rollout), so the worst case is a pure one-sided pump into the unique positive- g atom: $V(R) = \text{flip}(\hat{H}) + g_{\max}R$ exactly and $\tau^*(\alpha) = \alpha/g_{\max} - \hat{H}(\frac{1}{2})$ in closed form (tv_robustness_openr1_r483_result.json; scan match 10^{-9} , LP 10^{-6}). Since $g_{\max}$ tracks the certificates (0.0785/0.0460): $\tau^*(0.10) = \tau^*(0.05) = 0.168$ (halving α past 0.0785 halves $g_{\max}$, $0.25 \rightarrow 0.125$), while $\tau^*(0.02) = \tau^*(0.01) = 1$ — below the smallest certificate the frozen rule never stops, $V(R) \equiv 0$, no prior shift can break it. Non-monotonicity is thus maximal here ($0.168 \rightarrow 1$); zero- g mass 76.8% sits between OMR’s 36–47% and RLVE’s 74–87%, completing the pattern from the low-support side. *Conservation curve*. Appendix D maps $\tau^*(\alpha)$ on the dense continuum $\alpha \in [10^{-3}, 0.2]$ for all three carriers: the curve is analytically staircase-shaped — non-decreasing inside each constant-rule interval, so that every jump of τ^* is a strict *drop* exactly at a certificate threshold (analytic on OpenR1: τ^* drops by 0.314 at $\alpha = 0.0785$ and by 0.864 at 0.0460). The practical reading is a staircase interpretation of the certificate level: at a rule-change breakpoint, a small budget relaxation buys a discrete jump in certified drift tolerance, so α should be set on a step, not inside one.

Remark 3 (Continuous- α selection is free for BAYES-H). The stopping condition $c_{\hat{H}}(x, k) \leq \alpha$ does not involve α on the left-hand side, so the BAYES-H rule family is *nested* in α : as α varies continuously, the rule changes only at the finitely many breakpoints given by the distinct values of $c_{\hat{H}}(x, k)$ — at most $\sum_{k=3}^N (k+1)$ stoppable states (555 at $N=32$; 267 distinct values observed, plus the never-stop rule). These are three distinct counts: 555 stoppable states bound the possible rule changes; only 267 *distinct* certificate values are observed on the support, and the rule family has one rule per value *plus* the never-stop rule (268 in total; shrunk to 182 below for the tested range). A union bound over this breakpoint family therefore certifies the *entire continuum* $\alpha \in (0, 0.10]$ simultaneously — no pre-declared α grid is needed. The family can be shrunk further with no loss: for the range $\alpha \in (0, 0.10]$ only the 181 distinct positive certificate values ≤ 0.10 are rule-distinguishing (values > 0.10 never trigger any rule in the range; zero-certificate states stop under every $\alpha > 0$), so the exact effective family has 182 rules ($J = 6,501$ including the FIXED- k part) instead of the 9,339 obtained by counting all observed breakpoints; the four reference selections below are unchanged and their UCBs tighten by $\leq 6 \times 10^{-4}$ (prop_562bound_r479_result.json). On our calibration pool the enlarged family ($J_{\text{cont}} = 9,339$ vs. $J = 64$) inflates UCBs only through the logarithmic factor; the selected cost-optimal BAYES-H rule is bit-for-bit identical at $\alpha \in \{0.10, 0.05, 0.02\}$ and tightens only at $\alpha = 0.01$ (mean k $10.3 \rightarrow 11.9$), while the weaker FIXED-EB baseline loses its $\alpha = 0.02$ budget under *either* family size (alpha_continuous_r478_result.json).

Table 1: TEST readout ($n = 3607$ problems, $N = 32$; 8 rule-level rows under the Bonferroni CI of Section 5 plus FULL-32 as the zero-saving reference). Saving = $1 - \bar{k}/32$. BAYES-H attains the largest valid saving at $\alpha = 0.05$ and 0.02 ; at $\alpha = 0.10$ FIXED-EB saves 84.4% versus BAYES-H 84.0% (a 0.4-point gap in mean cost on this test pool, descriptive, no paired CI computed for this pair), while BAYES-H keeps the lowest realized flip at every level. WINDOW3 saves most but is invalid at $\alpha \leq 0.05$.

Rule	α	realized flip	mean k	saving
FULL-32	—	0	32.0	0
WINDOW3 (ESC-style)	—	0.0577	4.4	86.2%
FIXED-HOEF $k^*=7$	0.10	0.0645	7.0	78.1%
FIXED-EB $k^*=5$	0.10	0.0794	5.0	84.4%
BAYES-H	0.10	0.0370	5.1	84.0%
FIXED-HOEF $k^*=27$	0.05	0.0156	27.0	15.6%
FIXED-EB $k^*=17$	0.05	0.0324	17.0	46.9%
BAYES-H	0.05	0.0249	6.1	80.9%
FIXED-HOEF	0.02	<i>no certifiable budget</i>		
FIXED-EB $k^*=31$	0.02	0.0083	31.0	3.1%
BAYES-H	0.02	0.0091	8.5	73.5%

5 PROTOCOL: FIT/CAL/TEST WITH A LEGITIMATE SELECTION FAMILY

Split. 11,607 problems are partitioned once (seeded) into FIT (4000), CAL (4000), TEST (3607). FIT estimates $\hat{H}$ and freezes the BAYES-H certificate tables. CAL selects rules from the pre-declared family: fixed- k candidates $k \in \{3, 5, \dots, 31\} \times \alpha \in \{0.10, 0.05, 0.02, 0.01\}$ (60 certificates) and BAYES-H at the same 4 levels, with simultaneous empirical-Bernstein UCBs at per-test $\delta = 0.05/64$. $k^*(\alpha)$ is the smallest budget whose CAL UCB is $\leq \alpha$; BAYES-H is accepted at level α iff its CAL UCB is $\leq \alpha$. TEST is read exactly once. All evaluation uses the exact per-task flip functions ((1) for fixed- k ; the DP of Appendix B for BAYES-H), so TEST numbers are exact replay quantities, not MC estimates; the DP itself is cross-checked against replay MC (max deviation 0.043 within $3\sigma_{MC} = 0.075$).

Baselines. FULL- N ; FIXED-HOEF (fixed budget selected with distribution-free Hoeffding UCBs, the strongest model-free competitor); FIXED-EB (fixed budget with the same mixture-aware empirical-Bernstein machinery as BAYES-H — isolates the value of adaptivity); BAYES-UNIF (uniform prior; not a certificate when the prior is wrong); WINDOW3 (ESC-style: stop when 3 consecutive samples agree with the current majority — no certificate knob).

6 EXPERIMENTS

Carrier. OpenMathReasoning CoT shard 0 (Moshkov et al., 2025) (CC-BY-4.0): the `pass_rate_72b_tir` column gives, for 11,607 unique problems, the exact pass count K of Qwen2.5-Math-72B-TIR over $N = 32$ rollouts — no answer-extraction noise. Pass rates are strongly bimodal: 45% of problems have $p \leq 0.1$ or $p \geq 0.9$; 11.6% sit in $[0.4, 0.6]$. This carrier is a *count-exchangeable mixture* carrier: it supports the replay model exactly but contains no rollout order (Section 7).

Main table. Table 1 reports the single TEST readout (Bonferroni CI radius 0.0286 across the 9 reported rules). Every claim of “valid” means realized flip $\leq \alpha$; TEST is descriptive (validity was certified on CAL).

Fair comparison. Against the strongest distribution-free baseline (FIXED-HOEF) at the same α , BAYES-H’s paired per-problem saving gap is $+0.058$ (CI radius 0.029) at $\alpha = 0.10$ and $+0.652 \pm 0.029$ at $\alpha = 0.05$ — a $5.2\times$ improvement over the baseline’s 15.6% saving; at $\alpha = 0.02$ the baseline has *no* certifiable budget while BAYES-H still saves 73.5%. Against the strongest mixture-aware fixed budget (FIXED-EB), adaptivity adds $+34$ points of saving at $\alpha = 0.05$ (80.9% vs

46.9%) and +70 points at $\alpha = 0.02$ — the adaptive advantage grows as α tightens, because the conditional certificate stops early exactly on the bimodal mass (extreme- K problems stop at $k \approx 5$) and waits only on the 11.6% of mid-difficulty problems.

Cross-source transfer. Splitting CAL/TEST across the four AoPS problem sources (train on three, test on the fourth), BAYES-H stays valid on all transfers (flip 0.021–0.033 $< \alpha = 0.05$; saving 79–83%): moderate mixture shift preserves the certificate, as expected from the linearity of the population flip rate in H .

Second-shard replication and prior-mismatch transfer. We replicate the entire chain on a disjoint carrier, OpenMathReasoning CoT shard 1 (11,454 problems, same $N=32$ pass counts; SHA-256 pinned), whose mixture structure is near-identical (heterogeneity excess 0.125 vs 0.129; 44.4% extreme- p mass). Two runs share the CAL/TEST problems of shard 1 and differ only in the prior’s provenance. (i) *Within-shard*: the prior is fit on shard 1’s own FIT split; every qualitative conclusion reproduces — BAYES-H saves 80.6% at realized flip 0.025 ($\alpha = 0.05$) and 73.1% at 0.009 ($\alpha = 0.02$), vs 15.6% and *no certifiable budget* for FIXED-HOEF. (ii) *Cross-shard prior transfer*: the prior is fit on shard 0 FIT (total-variation distance to the shard 1 prior = 0.042) and the certificate is selected and read on shard 1. All numbers move by ≤ 0.4 points (saving 80.6%/73.2%; flips 0.025/0.009), confirming that the certificate is robust to realistic prior mismatch, as the linearity of the population flip rate in H predicts (Section 8).

Second model-carrier pair: OpenR1-Math-220k (DeepSeek-R1, $M = 2$). To probe the mechanism under a different model and a structurally opposite carrier, we run the identical chain on OpenR1-Math-220k (MIT; 9,374 unique problems, two deduplicated DeepSeek-R1 rollouts each, HF snapshot e4e141ec). The replay model instantiates exactly: the full vote is the 2-vote majority (ties broken by a pre-drawn fair coin), the only stoppable prefix is $k = 1$, and per-problem risk is analytic. The only identifiable prior at $M = 2$ is coarse (three atoms $p \in \{0, \frac{1}{2}, 1\}$, fit by exact pair-type moment matching): 25% always-fail, 23% coin-flip, 52% always-pass. Results (FIT/CAL/TEST 3000/3000/2853): (i) the conditional certificate attains the risk-cost *frontier*: at $\alpha = 0.20$ it stops on every problem (flip $0.059 \leq \alpha$, 50% saving); at $\alpha = 0.10$ full stopping has the tightest distribution-free UCB $0.080 \leq \alpha$ and is certified with the same realized flip and saving; at $\alpha = 0.05$ it stops only on the sub-population whose observed prefix certifies (positive-side prefix), saving 32% at flip $0.029 \leq \alpha$; at $\alpha = 0.02$ it *never* stops — and no legal certificate can, since stopping at $k=1$ on any problem carries posterior risk > 0.02 ; (ii) the realized flip of always stopping at $k=1$ (0.059) matches its analytic value $\mathbb{E}_H[p(1-p)] = 0.058$ to Monte-Carlo precision. Saving is bought by mixture *structure* that a prefix can resolve: with $M = 2$ there is at most one draw of prefix information, and the certificate extracts exactly the saving that structure allows — 50% at loose levels, partial at $\alpha = 0.05$, honestly nothing at $\alpha = 0.02$ — versus 81%/73% at $\alpha = 0.05/0.02$ on the $N=32$ OMR carrier. This is the cross-model evidence that the gains are mechanism-driven, not carrier luck.

Third model-carrier pair: RLVE (Qwen3-4B, $N = 8$). The two carriers above close the extremes ($N=32$ rich-prefix vs $M=2$ near-prefix-free). To probe the intermediate regime under a third model family and a different task family, we run the identical chain on RLVE teacher rollouts (Qwen3-4B-Thinking-2507, pass@8 on 9,000 verifiable counting/combinatorics environments; Apache-2.0; HF snapshot eaeec946). Success is the offline Gym-verifier reward being positive (a small mass of fractional partial credit is counted as failure). The mixture is again strongly U-shaped (42% extreme problems, heterogeneity excess 0.100 over the binomial baseline). Results (FIT/CAL/TEST 3000/3000/3000): (i) BAYES-H saves 57.6%/50.9%/48.2%/45.4% at $\alpha = 0.10/0.05/0.02/0.01$ with realized flips 0.031/0.010/0.005/0.002 — conservative at every level; (ii) the certificate passes at *every* level down to $\alpha = 0.01$, while *no* distribution-free budget is certifiable below $\alpha = 0.10$ (FIXED-EB and FIXED-Hoeffding both null at $\alpha \leq 0.05$) — at $\alpha = 0.10$ the paired saving gap over FIXED-Hoeffding is +0.20 (CI radius 0.03, significant); (iii) the exact DP prediction matches a 3,000-draw replay Monte-Carlo on held-out problems (flip within 0.004, mean k within 0.04). Two descriptive diagnostics: the generation-order (sample-id) success slope is flat (pooled slope -7.8×10^{-5} /trial; per-position rates 0.316–0.326), consistent with near-exchangeable sampling in this carrier; and the saving interpolates with resolvable struc-

ture exactly as the mechanism predicts (81% at $N=32$ with 15 stoppable budgets, 45–58% at $N=8$ with 3, 32–50% at $M=2$ with 1; all in-distribution readouts collected in Appendix E).

Why the window heuristic fails. On mid-difficulty problems ($K \approx 16$), three consecutive agreements with the current majority is a *common* event under near-coin-flip draws, not evidence the vote has settled: WINDOW3 realizes flip 0.058 at mean $k = 4.4$. Its flip rate is nearly flat in α (it has no certificate knob): it can only be tuned by window length, and its certified population UCB is 0.061 (`adaptive_sweep_r468_result.json`), so it barely misses even the loosest tested level $\alpha = 0.10$ on the full OMR test pool.

7 ORDERED DRIFT: FAILURE DOMAIN AND MARGIN REPAIR

The replay model assumes deployment prefixes are exchangeable with the held remainder. Real generation is ordered, so we stress-test every stopper under three *pre-fixed* drift mechanisms applied to the true counts ($\delta \in \{0, 0.05, 0.10, 0.15, 0.20\}$, 2000 TEST problems, 500 orders each): **E1** (front-load: the first k draws are enriched/depleted in passes by count $D = \lceil 2\delta \min(K, N-K) \rceil$), **E2** (linear per-draw pass-rate trend $p_j = K/N + \delta(2j/(N-1)-1)$), **E3** (adversarial block: the last $\lceil \delta N \rceil$ draws drift toward the minority side).

Findings (full tables in Appendix C): (i) all methods break under E1 by $\delta = 0.10$ – 0.15 ; the certificate is honest about its premise — E1 violates exchangeability by construction; (ii) under the gentler E2/E3, BAYES-H is the *most* robust stopper: at $\alpha = 0.10$ it remains valid to $\delta = 0.20$ under E2 (flip 0.083) while FIXED- k breaks at $\delta = 0.15$ and WINDOW3 breaks at $\delta = 0.20$ (flip 0.104; under E3 WINDOW3 stays valid throughout, its last-window rule never reaching the drifting tail); at $\alpha = 0.05$ under E3 it breaks only at $\delta = 0.15$ (flip 0.052), while FIXED-EB survives (it stops at $k = 17$, having seen half the block); (iii) validity under drift is bought by margin: stopping only when $c_H \leq \alpha - \gamma$ restores validity at a quantified saving cost. Under E3 at $\alpha = 0.05$, margin $\gamma = 0.025$ covers $\delta \leq 0.15$ (flip 0.0495, saving 77.2% vs 81.0% unmargined in the stress harness; the unmargined exact TEST readout of Table 1 is 80.9%); covering the extreme $\delta = 0.20$ adversarial block requires $\gamma > 0.03$ (saving cost > 7 points), and under the smoother E2 trend $\gamma = 0.03$ covers the full tested range $\delta \leq 0.20$ (flip 0.0491, saving 71.8%). The drift budget δ is estimable on CAL by trial-order regression when ordered rollouts *are* available; on this count-only carrier it must be assumed, which we state as a limitation.

8 THE MIXTURE IS THE OBJECT

Three converging lines of evidence motivate the mixture framework. (a) *Pooled correlation is heterogeneity*: on τ^2 -bench, fixed-effect demeaning collapses $\hat{\rho}$ from 0.5 to the $-1/(N-1)$ artifact floor (Remark 1). (b) *Ignoring the mixture undercovers*: a synthetic probit cluster study ($N \in \{10, 20, 40\}$, 2500 tasks/cell) shows i.i.d.-conformal stopping undercovers under heterogeneous pass rates (agreement $0.983 < 0.99$ target), while pooled- ρ corrections are valid but over-conservative. (c) *The plug-in mixture estimate undercovers too*: certifying with $\mathbb{E}_{\hat{H}}[f_K(k)]$ directly yields bootstrap p95 $0.051 > \alpha = 0.05$ at $k = 11$; the empirical-Bernstein layer is what restores finite-sample validity. Point-estimate correctness $\neq$ certificate calibration — this is the paper’s central methodological point.

9 LIMITATIONS

(i) The public carrier provides pass *counts*, not ordered rollouts: all main results are exact under the count-exchangeable replay model; ordered deployment validity is bounded only through the stress mechanisms of Section 7. The certificate is a replay-model conditional certificate, not an e-process: validity under arbitrary online stopping times in ordered deployment is outside the stated claims. (ii) The mixture prior is estimated from 4000 calibration problems of the same distribution; adversarial mixture shift beyond the tested transfer range can break the certificate; the exact critical radii of Proposition 1 quantify the tolerable shift ($\tau^* = 0.024$ – 0.157 across levels and shards) and show the failure boundary is sharp — at $\alpha = 0.01$ on shard 1 the observed cross-shard mismatch 0.042 already exceeds $\tau^*(0.01) = 0.024$; and the radii are *rule-conditional*: $\tau^*(\alpha)$ is a staircase

with jumps at the certificate thresholds (Appendix D), so the level α should be read as choosing a step of that staircase, not a point of a smooth trade-off. Along the estimation axis, Appendix D(e) shows τ^* is not monotone in the prior-fit size m ; the decomposition of Appendix D(f) attributes this to the rule the fit induces (freezing the prior leaves the non-monotonicity intact; freezing the rule removes the largest dips), not the prior’s sampling error. Appendix D(g) turns that attribution into a constructive repair: a per-atom flip budget implemented as a certificate-ordered subset of stopping states provably dominates the induced profile, restores the full simplex ($\tau^* = 1$, trivial-validity regime) at 70 of 72 cells for +1%–+11% extra votes, and repairs the coarse-fit breaks below — while two alternatives (prior support truncation, deadline truncation) provably do not. On OMR shard 1 the coarsest fit $m=125$ breaks the transferred certificate at $\alpha \leq 0.02$ while $m \geq 250$ — or the flip-budget repair at any m — restores it. Appendix E collects every in-distribution readout and its failure boundary in one matrix. (iii) Lemma 1’s monotonicity is proved in closed form (Appendix A) and independently verified exhaustively at $N = 32$. (iv) Main-table counts come from one model–task pair (Qwen2.5-Math-72B-TIR, math problems); generality across problem pools is supported by the disjoint second-shard replication and cross-shard prior transfer, and generality across models and task families by two further model–carrier pairs — DeepSeek-R1 rollouts on OpenR1-Math-220k ($M = 2$), on which the certificate attains its ordered risk-cost frontier, and Qwen3-4B rollouts on RLVE verifiable environments ($N = 8$), on which it is certified at every tested level and interpolates with the resolvable mixture structure (Section 6). (v) The forced-stop boundary at $k = N$ is absorbed into the population bound; conditional per-problem validity at forced stops is not guaranteed.

10 CONCLUSION

Treating across-task heterogeneity as the certificate object turns early stopping for majority voting into an exact conditional-inference problem: adaptivity is free (tower property), calibration is one empirical-Bernstein step over problems, and the result saves 81% of rollouts at certified flip level 0.05 where the strongest distribution-free baseline saves 16% and the strongest fixed mixture-aware budget saves 47%. The same framework admits an honest ordered-drift analysis with a margin repair, and a clear statement of what a count-only carrier can and cannot certify.

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A PROOFS

A.1 MONOTONICITY OF THE FIXED- k FLIP PROBABILITY (LEMMA 1)

Recall

$$f_K(k) = \mathbb{P}(\text{MV}(k) \neq \text{MV}(N) \mid K), \quad x_k \sim \text{HG}(N, K, k),$$

with the negative side winning ties ($\text{MV}(k) = 1$ iff $x_k > k/2$). We prove that $f_K(k)$ is non-increasing along the odd numbers $k \in \{1, 3, 5, \dots, N-1\}$ for every fixed K ; this suffices for the paper, since the fixed- k certificate family is evaluated on odd budgets only, and it makes $k^*(\alpha) = \min\{k : \mathbb{E}_H[f_K(k)] \leq \alpha\}$ well defined and binary-searchable for every mixture H (mixtures of nonincreasing functions are nonincreasing).

Setup and case split. $\text{MV}(k) \neq \text{MV}(N)$ can happen in only two ways:

- **Case U (upset).** $K \leq N/2$ (full vote negative) and $x_k > k/2$ (prefix votes positive).
- **Case R (reversal).** $K > N/2$ (full vote positive) and $x_k \leq k/2$ (prefix votes negative).

Write $f_K(k) = U_K(k) + R_K(k)$ with $U_K(k) = \mathbb{P}(x_k > k/2) \mathbf{1}\{K \leq N/2\}$ and $R_K(k) = \mathbb{P}(x_k \leq k/2) \mathbf{1}\{K > N/2\}$. At most one of the two terms is nonzero for any K .

The reflection identity. Swapping the labels “pass”/“fail” of all N rollouts maps the law $\text{HG}(N, K, k)$ of x_k onto the law of $k - x_k$ under count $N - K$, i.e. $\mathbb{P}_K(x_k > s) = \mathbb{P}_{N-K}(x_k < k - s)$. Hence, for odd k ,

$$U_K(k) = \mathbb{P}_K(x_k > k/2) = \mathbb{P}_{N-K}(x_k < k/2) = R_{N-K}(k) \quad (K \leq N/2),$$

since k odd makes $k/2$ a half-integer, so the strict and weak inequalities coincide. Every upset probability thus equals a reversal probability at the reflected count, and it suffices to prove

$$R_K(k+2) \leq R_K(k) \quad \text{for all } K > N/2 \text{ and odd } k, \quad (3)$$

which covers Case R directly and Case U by reflection (the same display applied at count $N - K$).

Coupling construction. Fix $K > N/2$ and odd $k \leq N - 3$. We couple the two hypergeometric draws on a single probability space. Let the population of N rollouts consist of K passes and $N - K$ fails. Draw an *ordered* sample of size $k + 2$ uniformly without replacement and split it as

$$\underbrace{X_1, \dots, X_k}_{\text{prefix } k} \mid \underbrace{X_{k+1}, X_{k+2}}_{\text{extra pair } (A, B)}.$$

The marginal law of $(X_1, \dots, X_k)$ is exactly $\text{HG}(N, K, k)$, and $x_{k+2} = x_k + A + B$ has law $\text{HG}(N, K, k+2)$. All statements below are about this joint space, so they hold for the two marginals simultaneously.

Let $t = k/2 + 1/2$ be the (integer) flip threshold: the prefix at k votes negative iff $x_k \leq t - 1$; the prefix at $k + 2$ votes negative iff $x_{k+2} \leq t$. Partition the event $\{x_k = s\}$ by the value of the pair

$A + B \in \{0, 1, 2\}$:

$$\begin{aligned} E_0 &= \{x_k = t - 1, A + B = 0\}, & \text{flip at } k, \text{ flip at } k + 2, \\ E_1 &= \{x_k = t - 1, A + B = 1\}, & \text{flip at } k, \text{ flip at } k + 2, \\ E_2 &= \{x_k = t - 1, A + B = 2\}, & \text{flip at } k, \text{ no flip at } k + 2, \\ F_0 &= \{x_k = t, A + B = 0\}, & \text{no flip at } k, \text{ flip at } k + 2, \\ F_1 &= \{x_k = t, A + B = 1\}, & \text{no flip at } k, \text{ no flip at } k + 2. \end{aligned}$$

All other states agree on both sides (e.g. $x_k \leq t - 2$ flips under both, since $A + B \leq 2$; $x_k \geq t + 1$ flips under neither). Therefore

$$R_K(k) - R_K(k + 2) = \mathbb{P}(E_2) - \mathbb{P}(F_0), \quad (4)$$

and it remains to show $\mathbb{P}(E_2) \geq \mathbb{P}(F_0)$.

Comparison of the two boundary atoms. Both atoms factor as (prefix probability) $\times$ (pair probability), and each factor is a ratio of binomial coefficients. With $k = 2t - 1$ the algebra closes completely. First,

$$\mathbb{P}(x_k = t - 1) = \frac{\binom{K}{t-1} \binom{N-K}{t}}{\binom{N}{k}}, \quad \mathbb{P}(x_k = t) = \frac{\binom{K}{t} \binom{N-K}{t-1}}{\binom{N}{k}},$$

since $k - (t - 1) = t$ and $k - t = t - 1$. Given $x_k = t - 1$, the remainder holds $K - t + 1$ passes out of $N - k$, so $\mathbb{P}(A + B = 2 \mid x_k = t - 1) = \binom{K-t+1}{2} / \binom{N-k}{2}$; given $x_k = t$, the remainder holds $N - K - t + 1$ fails, so $\mathbb{P}(A + B = 0 \mid x_k = t) = \binom{N-K-t+1}{2} / \binom{N-k}{2}$. The denominators $\binom{N}{k} \binom{N-k}{2}$ are common to both atoms, hence

$$\frac{\mathbb{P}(E_2)}{\mathbb{P}(F_0)} = \frac{\binom{K}{t-1} \binom{N-K}{t} \binom{K-t+1}{2}}{\binom{K}{t} \binom{N-K}{t-1} \binom{N-K-t+1}{2}} = \frac{K - t}{N - K - t},$$

where the last equality uses the three elementary ratios $\binom{K}{t-1} / \binom{K}{t} = \frac{t}{K-t+1}$, $\binom{N-K}{t} / \binom{N-K}{t-1} = \frac{N-K-t+1}{t}$, and $\binom{K-t+1}{2} / \binom{N-K-t+1}{2} = \frac{(K-t+1)(K-t)}{(N-K-t+1)(N-K-t)}$, whose product telescopes to the displayed ratio. Therefore, with $C := \binom{K}{t} \binom{N-K}{t-1} \binom{N-K-t+1}{2} / [\binom{N}{k} \binom{N-k}{2}] \geq 0$,

$$\mathbb{P}(E_2) - \mathbb{P}(F_0) = C[(K - t) - (N - K - t)] = C(2K - N) \geq 0, \quad (5)$$

because $K > N/2$ in Case R (and if $t > K$ then $\mathbb{P}(x_k = t - 1) = 0$, so both atoms vanish). By (4) this proves (3) and, by the reflection identity, $U_K(k + 2) \leq U_K(k)$ as well, completing the proof of monotonicity along odd k . $\square$

Remark 4 (Verification at $N = 32$). Independently of the proof, the monotonicity was verified by exhaustive enumeration over all 33×15 (K, k) cells at $N = 32$ (zero violations, tolerance 10^{-12} ; artifact L2_lemmas_r468_result.json, odd- k grid $\{3, 5, \dots, 31\}$).

Remark 5 (Why only odd k enters the certificate). Requiring k odd removes tie-breaking from the certificate family (every prefix has a strict majority side), and the binary search for $k^*(\alpha)$ is over the odd ladder only. Nothing else in the paper uses monotonicity at even k .

A.2 FORCED-STOP BOOKKEEPING (THEOREM 1)

The stopping rule is $\tau = \min\{k \geq 3 : c_H(x_k, k) \leq \alpha\}$, truncated at $k = N$ (forced stop). The tower argument of Theorem 1 applies to the truncated rule verbatim: τ is still a stopping time with respect to the prefix filtration, and

$$\mathbb{P}(\text{flip}) = \mathbb{E}[c_H(x_\tau, \tau)] = \mathbb{E}[c_H \mathbf{1}\{\tau < N\}] + \mathbb{E}[c_H \mathbf{1}\{\tau = N\}].$$

The first term is bounded by $\alpha \mathbb{P}(\tau < N) \leq \alpha$ pointwise, because the rule stops only at states with $c_H \leq \alpha$. The second term is *not* bounded by α pointwise: at a forced stop the prefix certificate may exceed α (and can be as large as the posterior mass on the opposite side). This is precisely why the finite-sample guarantee is stated at the *population* level (Theorem 2): the exact per-task flip probability $g(K; \alpha, \hat{H}_m)$ computed by the DP of Appendix B includes the forced-stop contribution $\mathbb{E}[c_H \mathbf{1}\{\tau = N\} \mid K]$ exactly, and the empirical-Bernstein layer then bounds the population mean $\sum_K H[K]g(K)$. Conditional per-problem validity at forced stops is not claimed (Limitation (v)).

Remark 6 (The forced-stop mass is small in practice). On the TEST split, the exact fraction of paths reaching $k = N$ under BAYES-H is 0.59% at $\alpha = 0.05$ and 1.10% at $\alpha = 0.02$ (the certificate stops early exactly on the bimodal mass); their contribution is included in g exactly regardless (Appendix B).

A.3 LINEARITY OF THE POPULATION FLIP RATE

For fixed rule and fixed prior $\hat{H}_m$ used *inside* the certificate, the per-task flip probability $g(K)$ depends only on the true count K (via the hypergeometric law of prefixes) and on $\hat{H}_m$ (through the posterior threshold), but is a fixed function once $\hat{H}_m$ is frozen after FIT. The population replay flip rate is therefore

$$\mathbb{P}_{\text{replay}, H}(\text{flip}) = \sum_{K=0}^N H[K] g(K),$$

a *linear* functional of the true mixture H . Consequences used in the main text: (i) no Jensen gap — bounding $\bar{g}_{\text{CAL}}$ by empirical Bernstein bounds the population quantity directly, with estimation error of $\hat{H}_m$ entering only through the empirical mean of a bounded per-task quantity $g \in [0, 1]$; (ii) graceful degradation — if the deployment mixture shifts from H to H' , $|\mathbb{P}_{H'} - \mathbb{P}_H| \leq \|H' - H\|_1$ since $g \in [0, 1]$; (iii) Bonferroni over the pre-declared family of $J = 64$ rules legitimizes the min- k selection, because simultaneous UCBs on all J linear functionals imply the UCB of the selected rule.

B EXACT DYNAMIC PROGRAM FOR THE ADAPTIVE RULE

We compute, for a fixed true count K and frozen prior $\hat{H} = \hat{H}_m$, the exact per-task flip probability $g(K; \alpha, \hat{H})$ and expected stopping time $\mathbb{E}[\tau \mid K]$ of BAYES-H under the replay model, by dynamic programming over prefix states.

State space and transitions. A state is a prefix (k, x) with $0 \leq x \leq k \leq N$, $k \geq 0$. Under count K , the next draw is a uniform element of the remainder, so the transition kernel is

$$\mathbb{P}(x_{k+1} = x + 1 \mid k, x, K) = \frac{K - x}{N - k}, \quad \mathbb{P}(x_{k+1} = x \mid k, x, K) = \frac{N - K - (k - x)}{N - k},$$

and the initial state has $\mathbb{P}(x_0 = 0) = 1$. The state graph is a DAG ($O(N^2)$ states, $O(N^2)$ edges), so a single backward pass is linear in the number of states.

Certificate table. The posterior $\mathbb{P}(K = j \mid k, x) \propto \hat{H}[j] \text{HG}(N, j, k, x)$ and $c_H(x, k)$ are pre-computed once for all (k, x) on the $O(N^2)$ grid (the forced-stop value at $k = N$ is the indicator $\mathbf{1}\{\text{side}(x, N) \neq \text{side}(K, N)\}$, deterministic in K). A state is a *stop state* iff $k \geq 3$ and $c_H(x, k) \leq \alpha$; $k = N$ is a stop state by truncation.

Recursions. For a fixed K , define $V(k, x) = \mathbb{P}(\text{MV}(\tau) \neq \text{MV}(N) \mid k, x, K)$ and $T(k, x) = \mathbb{E}[\tau - k \mid k, x, K]$. Then

$$V(k, x) = \begin{cases} \mathbf{1}\{\text{side}(x, k) \neq \text{side}(K, N)\}, & (k, x) \text{ stop state,} \\ \frac{K - x}{N - k} V(k + 1, x + 1) + \frac{N - K - k + x}{N - k} V(k + 1, x), & \text{otherwise,} \end{cases}$$

and the same recursion with $T(\cdot) = 0$ at stop states and $T(k, x) = 1 + (\text{continuation mean})$ otherwise. The requested quantities are $g(K) = V(0, 0)$ and $\mathbb{E}[\tau \mid K] = T(0, 0)$. The per-task exact saving is $1 - \mathbb{E}[\tau \mid K]/N$; population quantities follow by linearity, $\sum_K H[K]g(K)$, for any mixture H . The same forward pass yields the exact forced-stop occupancy $\mathbb{P}(\tau = N \mid K)$: on the TEST split it is 0.59% at $\alpha = 0.05$ and 1.10% at $\alpha = 0.02$.

Complexity. For each K : $O(N^2)$ states, $O(1)$ work per state after the certificate table is frozen, so all 33 values of K cost $O(N^3) \approx 3.6 \times 10^4$ flops at $N = 32$ — negligible. This is why every TEST number in the main table is an *exact replay quantity*, not a Monte Carlo estimate.

Cross-check against simulation. The DP is validated against an independent replay Monte Carlo implementation (fresh random prefixes) on the same frozen prior: the maximum absolute deviation over the checked cells is 0.043, within the pre-set tolerance $3\sigma_{MC} = 0.075$ (the check includes cells at $\alpha = 0.01$, whose realized flips are small and inflate the pooled MC standard error); PASS. See `fit_cal_test_r469.py` and the self-check block of `fit_cal_test_r469_result.json`.

C FULL DRIFT TABLES

Tables 2 and 3 report the complete drift stress grid behind Section 7 (`drift_stress_r469_result.json`; 2000 TEST problems, 500 orders per cell). Under the smooth mechanisms E2/E3, BAYES-H is the most robust stopper at both levels; under the premise-breaking E1 front-load all certificates fail together, as they should.

Table 2: Full drift stress table at $\alpha = 0.05$ (2000 TEST problems, 500 orders each; flip in %, mean budget k in parentheses). **Bold:** realized flip exceeds α (certificate broken).

Mechanism	δ	BAYES-H	FIXED-EB	FIXED-HOEF	WINDOW3
E1 (front-load)	0.0	2.5 (6.1)	3.2 (17.0)	1.6 (27.0)	5.8 (4.4)
	0.05	4.8 (6.1)	4.9 (17.0)	4.1 (27.0)	7.1 (4.4)
	0.1	7.6 (6.2)	8.2 (17.0)	7.6 (27.0)	9.6 (4.3)
	0.15	11.4 (6.0)	12.1 (17.0)	12.4 (27.0)	12.2 (4.2)
	0.2	13.0 (5.5)	13.7 (17.0)	13.6 (27.0)	13.5 (4.1)
E2 (linear trend)	0.0	4.7 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.05	4.8 (6.2)	4.7 (17.0)	3.7 (27.0)	7.0 (4.4)
	0.1	5.2 (6.3)	4.9 (17.0)	3.7 (27.0)	7.6 (4.5)
	0.15	5.9 (6.4)	5.3 (17.0)	3.7 (27.0)	8.8 (4.6)
	0.2	7.0 (6.5)	6.0 (17.0)	3.7 (27.0)	10.4 (4.6)
E3 (adversarial block)	0.0	4.7 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.05	4.8 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.1	4.9 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.15	5.2 (6.1)	4.7 (17.0)	3.8 (27.0)	6.8 (4.4)
	0.2	5.5 (6.1)	4.7 (17.0)	4.2 (27.0)	6.8 (4.4)

Table 3: Full drift stress table at $\alpha = 0.1$ (2000 TEST problems, 500 orders each; flip in %, mean budget k in parentheses). **Bold:** realized flip exceeds α (certificate broken).

Mechanism	δ	BAYES-H	FIXED-EB	FIXED-HOEF	WINDOW3
E1 (front-load)	0.0	3.7 (5.1)	7.9 (5.0)	6.4 (7.0)	5.8 (4.4)
	0.05	5.6 (5.1)	9.1 (5.0)	7.8 (7.0)	7.1 (4.4)
	0.1	8.4 (5.2)	11.0 (5.0)	10.1 (7.0)	9.6 (4.3)
	0.15	11.9 (5.0)	12.7 (5.0)	12.3 (7.0)	12.2 (4.2)
	0.2	13.3 (4.7)	13.7 (5.0)	13.5 (7.0)	13.5 (4.1)
E2 (linear trend)	0.0	5.4 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.05	5.5 (5.2)	8.9 (5.0)	7.5 (7.0)	7.0 (4.4)
	0.1	6.0 (5.3)	10.0 (5.0)	8.3 (7.0)	7.6 (4.5)
	0.15	6.9 (5.3)	11.6 (5.0)	9.7 (7.0)	8.8 (4.6)
	0.2	8.3 (5.4)	13.7 (5.0)	11.6 (7.0)	10.4 (4.6)
E3 (adversarial block)	0.0	5.4 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.05	5.4 (5.1)	8.6 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.1	5.5 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.15	5.6 (5.1)	8.7 (5.0)	7.4 (7.0)	6.8 (4.4)
	0.2	5.7 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)

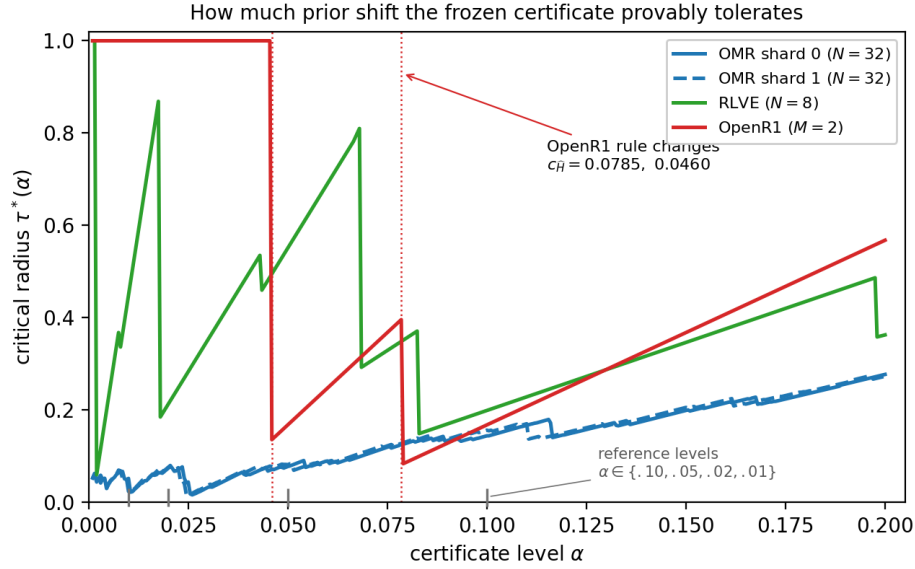


Figure 2: Critical radius $\tau^*(\alpha)$ of Proposition 1 on the continuum $\alpha \in [10^{-3}, 0.2]$ (step 5×10^{-4}), computed exactly per level from the frozen g -profiles of the three carriers (OMR both FIT shards; OpenR1 exact in closed form). Grey tick marks: the four reference levels; red dotted lines: the OpenR1 certificate thresholds 0.0785/0.0460 where the frozen rule changes. Artifact: `tv_conservation_r484_result.json`.

D THE CRITICAL-RADIUS CONSERVATION CURVE

Figure 2 makes the rule-change structure of the certificate visible. (a) *Staircase law*. Within any α -interval on which the frozen rule is unchanged, the whole vector g — hence $V(R)$ — is constant, so $\tau^*(\alpha) = \max\{R : V(R) \leq \alpha\}$ is *non-decreasing*; every strict decrease of τ^* coincides with a rule-change breakpoint (verified at every grid cell of every carrier; the rule-change counts on the plotted range are 96/93/7/2 for OMR shard 0 / OMR shard 1 / RLVE / OpenR1). (b) *Analytic breakpoints on OpenR1*. With $V(R) = \text{flip}(\hat{H}) + g_{\max}R$ exactly, $\tau^*(\alpha) = \alpha/g_{\max} - \hat{H}(\frac{1}{2})$: at $\alpha = 0.07852$ the fail-prefix stop engages and $g_{\max}$ doubles ($0.125 \rightarrow 0.25$), so τ^* jumps down $0.396 \rightarrow 0.082$ ($\Delta = -0.314$); at $\alpha = 0.04598$ the pass-prefix stop disengages and the frozen rule degenerates to never-stop ($V(R) \equiv 0$ for any prior), so the whole-simplex plateau $\tau^* = 1$ holds below this threshold and τ^* jumps down $1 \rightarrow 0.136$ ($\Delta = -0.864$) as α crosses it upward. (c) *Whole-simplex plateaus*. Wherever the frozen rule’s worst case cannot exceed α for any prior ($\max_K g(K) \leq \alpha$), $\tau^* = 1$: RLVE for $\alpha < 0.002$ and OpenR1 for $\alpha < 0.0460$; OMR’s rule never stops only below 10^{-3} , off the plotted range. (d) *Alignment co-variation (decomposed)*. The zero- g alignment number co-varies with τ^* at pooled Spearman 0.75 across the 16 carrier $\times$ level cells (descriptive; the per-cell alignment is itself α -dependent). The decomposition audit (`spearman_decomposition_r486_result.json`) shows this is a between-carrier law, not a within-carrier one: 1.0 across the four carrier means, 0.63 on the 14 non-plateau cells, 0.44–0.90 under leave-one-carrier-out; within-carrier it is $-0.95/-0.89$ (OMR shards), $+0.32$ (RLVE), $+1.0$ (OpenR1) — the OMR reversal is the staircase mechanism itself (level tightening adds zero- g mass on the lowest- g atoms only). Heterogeneity is not a confounder: het-excess vs τ^* Spearman 0.01, het-excess vs alignment 0.05 (16 cells). *Practical reading*. The staircase shape is a design signal for choosing the certificate level: inside a constant-rule interval, loosening α buys certified drift tolerance gradually; at a breakpoint it buys (or loses) a discrete jump — α should be set on a step, not inside one. (e) *How much FIT data buys how much robustness*. Rebuilding the prior from nested prefixes of the frozen FIT order at $m \in \{\frac{1}{32}, \dots, 1\} \times m_{\text{full}}$ and recomputing $\tau_m^*(\alpha)$ exactly (`prior_fit_size_r491_result.json`) gives an operational size rule. On OMR the $\tau_m^*(0.05)$ curves already clear the observed cross-shard transfer 0.042 at the smallest tested pool $m=125$ (shard 0: 0.062; shard 1: 0.046), i.e. cross-shard robustness does *not* require the full 4000-

problem fit. The radius is *not* monotone in m (OMR shard 1 at $\alpha = 0.01$: 0.010 at $m=125$, up to 0.056 at $m=1000$, back to 0.024 at $m=4000$): growing the fit moves both the prior (better) and the frozen rule (larger $g_{\max}$ possible), and the rule channel can dominate — τ_m^* is a property of the fitted pair $(\hat{H}_m, \text{rule}_m)$, not of the sample size alone. The closed loop of Proposition 1 holds at every one of the 51 cells with $\text{TV}(\hat{H}_m, \hat{H}_{\text{full}}) \leq \tau_m^*$: the m -rule’s exact flip under the full-data prior never exceeds α (0 violations), while at the coarsest OMR shard 1 fit ($m=125$, TV 0.175, outside the ball) the transferred rule genuinely breaks (flip 0.0244 > 0.02, 0.0165 > 0.01) and is repaired from $m=250$ onward — inside/outside the radius predicts validity/failure exactly. The estimation error itself scales as $\text{TV}_m \approx C/\sqrt{m}$ (within-carrier spread of $\text{TV}_m\sqrt{m}$ below $1.5\times$ on OMR; the RLVE spread reaches $3.5\times$ only because its coarsest prefixes carry a missing-mode error $\sim \sqrt{m_{\text{full}}/m}$ larger). (f) *Which channel moves τ_m^* : rule, not prior — with one caveat.* We decompose the m -sensitivity into its two moving parts by computing, at every (carrier, m , α) cell, the two counterfactual radii (rule_channel_r494_result.json): $\tau_m^{*\text{pf}} = \tau^*(\hat{H}_{\text{full}}, g_m)$ freezes the prior at the full fit and lets only the rule vary with m , and $\tau_m^{*\text{rf}} = \tau^*(\hat{H}_m, g_{\text{full}})$ freezes the rule and lets only the prior vary. The recomputed both-moving curve reproduces the m -grid of prior_fit_size_r491_result.json bit-exactly (0/72 mismatches). Three findings. (i) *The rule channel drives the non-monotonicity.* With the prior frozen at $\hat{H}_{\text{full}}$, $\tau_m^{*\text{pf}}$ is still non-monotone at OMR shard 1 $\alpha=0.01$: 0.000 at $m=125$ rising to 0.060 at $m=1000$ then falling to 0.024 at $m=4000$ — the same up-then-down shape as the both-moving curve — because the refitted rule’s $g_{\max}$ itself varies non-monotonically (0.563 $\rightarrow$ 0.075 $\rightarrow$ 0.098) as the coarse prefix shifts which budgets are certifiable. (ii) *The estimation channel is benign on average but not uniformly.* Holding the rule fixed, a coarser prior can *shrink* the radius below the full-fit value in 31 of 72 cells (all on OMR shard 1 and RLVE; largest deficit 0.053 at RLVE $m=93$, $\alpha=0.05$). The mechanism is the base-flip term $\sum_K \hat{H}[K] g(K)$: a coarse prefix that under-covers the likely-correct (low- g) region and over-weights high- g mass raises the base flip, and — because the *frozen full rule* already has large g exactly on those atoms — the certificate has less slack to spend on drift. This is the missing-mode failure of (e) seen through the rule: the same coarse prefix that carries a $\sim \sqrt{m_{\text{full}}/m}$ estimation error also pays it on the worst (largest- g) coordinates. (iii) *The RLVE coarsest dip is pure rule-channel.* The collapse of the both-moving radius at RLVE $m=187$, $\alpha=0.05$ (0.235, far below both $m=93$ at 0.616 and $m=375$ at 0.534) survives with the prior frozen ($\tau_{187}^{*\text{pf}} = 0.222$) and vanishes with the rule frozen ($\tau_{187}^{*\text{rf}} = 0.530$): it is entirely a $g_{\max}$ event (0.179 vs. 0.071 for the full rule), not an estimation one. *Reading.* The robustness certificate is governed by the *rule* the fit induces more than by the prior’s sampling error: two fits of the same size can certify very different radii because the induced rule differs, and the only lever that reliably buys radius is one that shrinks the induced g -profile (e.g. the margin repair of Section 7), not one that merely adds FIT data. (g) *A constructive repair that buys radius: flip-budget state subsets.* The reading of (f) — shrink the induced g -profile — is here made executable (full_sweep_r498e_result.json). Fix a per-atom flip budget c . On each atom K , order the original rule’s stopping states by their certificate value and keep the longest prefix whose cumulative flip contribution is at most c ; states not kept run to n . The realized subset rule satisfies, pointwise on every atom, $g_S(K) \leq \min(g_{\text{orig}}(K), c)$ (domination), its stopping set is a subset of the original rule’s (it never stops earlier, so the deployment cost is only extra votes), and its exact radius is recomputed by the same linear program — nothing is re-estimated from data. Sweeping $c \in \{0.30, \dots, 0.01\}$ over all 72 carrier $\times$ m $\times$ α cells of (e): recomputation anchors the grid bit-exactly (0/72 mismatches), domination holds at every atom of every cell, and at the best cap every cell reaches $\tau^* = 1$: for 70 of 72 cells this is a strict gain over the original radius at +1% to +11% mean votes (max +11.3% at OMR shard 0 $m=500$, $\alpha=0.10$; RLVE never exceeds +3.3%), while the remaining two (RLVE at $\alpha=0.02$) are already whole-simplex under the original rule and the sweep selects no cap. The $\tau^* = 1$ -by-cap cells sit in the *trivial-validity* regime: the capped profile satisfies $\max_K g_S(K) \leq \alpha$ and $\text{bases}_S \leq \alpha$ by construction, so the certificate survives any bounded prior shift. This is not “infinite robustness for free”: the bill is paid by the realized base-flip level bases_S (e.g. 0.0054 $\ll$ $\alpha=0.05$ at OMR $m=4000$, cap 0.02) and by the extra votes, both reported per cell. The pitfall cell of (f) is repaired exactly: RLVE $m=187$, $\alpha=0.05$ goes from $\tau^* = 0.235$ to 1.0 at cap 0.07 with +0.5% votes; the coarse OMR shard 1 $m=125$ fits all exceed half their full-fit radius at every level (e.g. $\alpha=0.01$: 0.0095 $\rightarrow$ 1.0). Two negative results delimit the mechanism. First, *support truncation fails*: zeroing the low-mass atoms of the prior and renormalizing changes the induced certificate not at all (bit-identical g -profiles, 0 differences in 6 probed cells $\times$ 5 truncation levels; support_trunc_repair_r498_result.json) — the spikes of (f) live on near-zero-

mass boundary atoms that the base term never visits but the ℓ_1 adversary targets, and removing them from the prior does not touch the local likelihood geometry that induces them. Second, *deadline truncation fails*: capping each atom's stopping time at $\kappa(K)$ is non-monotone in κ (flip 0.014 at $\kappa=8$ rising to 0.357 at $\kappa=7$ and 0.5 at $\kappa=5$ on a probed RLVE atom) and can certify *less* than the original rule (RLVE $m=187$, $\alpha=0.05$: $0.235 \rightarrow 0.133$; `efficiency_r498c.result.json`) — a hard deadline deletes informative late states wholesale, whereas the certificate-order subset deletes exactly the states that are expensive to certify. Together with (f) this closes the loop: moving the prior does nothing, moving the time axis hurts, moving the *stopping-state set* in certificate order repairs — the rule channel is the lever, and the effective control is a flip budget, not a data budget. The repair is also not a degenerate retreat to stopping at the first informative prefix: a fixed $k=3$ stopper on OMR shard 1 $m=125$ has base flip 0.093 ($9\times$ above $\alpha=0.01$, $\tau^*=0$) against the repaired rule's $\text{base}_S = 0.0036$. *Universal budget*. The sweep also pins down a single carrier- and m -independent repair budget (`universal_cap_r500.result.json`): at cap 0.01 — the smallest level of the paper's grid — *every* one of the 72 cells reaches $\tau^*=1$, at $+1.0\%$ to $+22.3\%$ mean votes (OMR; $\text{RLVE} \leq +5.2\%$) and realized base flip ≤ 0.0038 . The boundary is sharp within the scanned grid: at cap 0.015 exactly the 18 cells with $\alpha=0.01$ (6 per carrier, every m) still fall short of the whole simplex, while all 54 cells with $\alpha \geq 0.02$ are already repaired. A budget at or below the smallest level of interest therefore repairs every fit size and every carrier we tested, and no smaller scanned budget suffices for the strictest level — the repair law is a property of the cap relative to α , not of m or the carrier. A fine-scale pass (`critical_cap_r503.result.json`; caps at 5×10^{-4} resolution plus bisection to $\approx 6 \times 10^{-7}$ on the full-radius prefix) sharpens this into an exact edge law. Because domination gives $\max_K g_S(K) \leq c$ pointwise, every cap $c \leq \alpha$ certifies the whole simplex mechanically, so the effective critical cap $c^* = \sup\{c : \tau^*(c) = 1\}$ satisfies $c^* \geq \alpha$ in every cell; the supremum is attained strictly above α whenever the realized staircase leaves $\max_K g_S(K) \leq \alpha$ beyond $c = \alpha$ — which happens in all 72 cells. The margin above α is carrier-resolved: it stays $\leq +0.8\%$ at $\alpha \geq 0.02$ on OMR (one shard 1 cell excepted, $+2.8\%$) and reaches $+10.6\%$ at $\alpha=0.01$ on OMR (fine $N=32$ certificate geometry). On RLVE the margin is not just measured but *derived* (`edge_law_r504.result.json`): the flip mass of stopping at state (k, x) on an atom is the hypergeometric reach probability $\binom{K}{x} \binom{n-K}{k-x} / \binom{n}{k}$, a rational number that does not depend on the fitted prior; the smallest certificate-greedy achievable flip sum above α is thus computable from the certificate table alone, and it equals the scan-measured edge wherever the scan bracketed one (all 18 such cells). On RLVE ($N=8$) this quantum sits at $c^*=3/28$ at $\alpha=0.10$ ($15/14 \cdot \alpha$), $1/14$ at $\alpha=0.05$ ($10/7 \cdot \alpha$), and $1/70$ at $\alpha=0.01$ ($10/7 \cdot \alpha$), identical at every one of the six fit sizes — the edge is combinatorics of the binding atom, not of the prior. At $\alpha=0.02$ the cell splits by a certificate straddle: at the four fit sizes whose binding state $(4, 1)$ has certificate ≤ 0.02 the edge is $1/14$ ($25/7 \cdot \alpha$, beyond the scanned band, which had reported only the censored bound $c^* \geq 2\alpha$), while at $m \in \{93, 750\}$ that state's certificate crosses $\alpha=0.02$ under the coarse-prefix prior, every achievable flip sum stays $\leq \alpha$, and the full radius survives *every* budget ($\tau^*=1$ certified at cap 0.5, $\max_K g_S=1/56$). On OMR ($N=32$) no such $O(\alpha)$ gap exists: the per-atom closed-form edge upper-bounds the budget-constrained scan edge (equality at 11 of 48 cells) and stays within $+10.6\%$ of α in every cell — the quantization of the repair edge is a small- N phenomenon. A tightness audit of the bound (`edge_tightness_r505.result.json`) separates the 48 OMR cells into two regimes by the sign of the certified slack one step below the closed edge: in 11 cells the realized $\max_K g_S$ at cap $c_{\text{closed}}^* - \epsilon$ is still $\leq \alpha$ (the binding atom's crossing state has not yet entered), so the scan edge and the closed edge coincide up to the scan's bracket resolution — the closed form is tight; in the remaining 37 cells the realized $\max_K g_S$ at $c_{\text{closed}}^* - \epsilon$ already exceeds α by 2.9×10^{-5} to 3.7×10^{-3} , so the budget-constrained greedy must drop the crossing prefix and $c_{\text{scan}}^* < c_{\text{closed}}^*$ strictly. The two regimes are disjoint in sign (nearest slacks -2.43×10^{-5} and $+2.91 \times 10^{-5}$, an inner margin ratio of $1.20\times$ — correcting the $\geq 2.4\times$ printed in v5.19–v5.20), so tightness is decided by a sign, not a fitted threshold. (This audit also corrected the equality count reported above in v5.18 from 12 to 11: the twelfth cell, shard 1 $m=2000$ $\alpha=0.10$, differs by 8.2×10^{-5} — two orders of magnitude above the scan's rounding noise — and had been absorbed by the verifier's 10^{-4} tolerance.) A discriminating audit at sub-artifact resolution (`discriminant_r507.result.json`) head-to-heads the slack-sign classifier against the two falsified first-draft conditions (recorded in `edge_tightness_r505.result.json`) on all 48 cells: the sign rule is exact (11 TP / 37 TN, no errors, invariant to the probe offset over $[5 \times 10^{-8}, 10^{-6}]$); the unique-binding-atom biconditional errs on 22 cells (19 FP +3 FN); and the best fixed margin threshold achieves only the always-strict constant (11 errors). The audit also maps the tolerance window itself: the 11-cell tight count is stable

on $[5 \times 10^{-7}, 2 \times 10^{-6}]$, degenerates to 3 at 2.5×10^{-7} (the scan edge’s six-decimal storage spreads exact equalities to $\pm 4 \times 10^{-7}$), and would absorb the genuine twelfth cell only above 8.22×10^{-5} — the usable discriminating band is $[5 \times 10^{-7}, 8.22 \times 10^{-5}]$, floored by the artifact’s own rounding resolution. Under $\pm 5 \times 10^{-7}$ scan perturbation (worst-case rounding) the sign rule and the scan verdict agree on all 48 cells; at exactly one bisection step ($\pm 6 \times 10^{-7}$) a single boundary cell flips per side, and no strict cell is ever absorbed. The universal $c \leq \min \alpha$ rule is thus the carrier-independent safe budget, and any budget inside the derived margin above α is free extra stopping mass, not a risk. *One geometry* (discrete_geometry_r506_result.json). All three non-monotone phenomena of this section are one object, the discrete stopping-state set: $\tau^*(\text{cap})$ is *not* a monotone function (323 of 792 adjacent-cap pairs on the coarse 12-cap grid violate monotonicity; the r501 working note misprinted the pair count), because both scalarizations it reads jump with the certificate-ordered prefix. Yet its full-radius crossing set is always a *prefix*: on the fine grid, every cap at or below c^* certifies the whole simplex and no cap above c^* does (0 holes and 0 overhangs across all 72 cells). The reconciliation is structural, not curve-fitting: shrinking the cap excludes a certificate-ascending prefix, so both scalars $\max_K g_S$ and base_S are monotone in the cap (0 and 0 violations respectively); once $\tau^*=1$ both lie at or below α and must stay there, while inside the sub-unit region the two scalarizations disagree and the staircase interior is free to wobble. The dead-line’s flip sequence is the same object read along the time axis (0.014, 0.357, 0.214, 0.5, 0.243, 0.5 as κ shrinks on the probed RLVE atom), and the closed-form edge is the same object read exactly: the hypergeometric flip masses are the quantum, the prefix structure gives the edge c^* , and the slack sign at one step below the closed edge decides its tightness. Reading any of the three as a smooth dial invites exactly the three failures documented above; reading it as a discrete set gives the safe budget, the derived edge, and the two operating ends (cheapest marginal repair at +0.09%–+1.29% votes; universal repair at +1.05%–+16.9%) of one rule.

Proposition 2 (Prefix law and two-scalar monotonicity of the full-radius crossing set). *Fix a cell $(\hat{H}_m, \alpha)$ and let $g_{S(c)}$ be the realized profile of the certificate-ordered state-subset rule at flip budget c , with scalarizations $\bar{g}(c) = \max_K g_{S(c)}(K)$ and $\text{base}_S(c) = \sum_K \hat{H}_m[K] g_{S(c)}(K)$, and write $\tau(c) = \tau^*(\hat{H}_m, g_{S(c)}; \alpha)$. Then: (i) both scalarizations are nondecreasing in c ; (ii) for every c , $\tau(c) = 1$ if and only if $\text{base}_S(c) \leq \alpha$ and $\bar{g}(c) \leq \alpha$; (iii) the full-radius crossing set $\{c : \tau(c) = 1\}$ is downward closed — a prefix — while $\tau(c)$ itself is free to be non-monotone inside the sub-unit region; and the crossing out of $\tau=1$ as the budget grows is base_S -driven: below α both conditions of (ii) already hold at $c = \alpha$ (by domination $\bar{g}(\alpha) \leq \alpha$) and persist for all $c < \alpha$ by (i), so at $c > \alpha$ only the $\text{base}_S(c) \leq \alpha$ condition can bind.*

Proof. (i) With nondecreasing budget, the greedy keep rule retains a nested family of kept sets: every state kept at c' is also kept at $c > c'$ — the certificate-sorted scan reaches that state with at least as much remaining budget, and its own contribution is unchanged. Hence $S(c') \subseteq S(c)$, and since each kept state’s flip contribution is nonnegative, both $\bar{g}$ and base_S are nondecreasing. (ii) By Proposition 1 the two-sided greedy over B_R adds mass at the highest- g atom and drains it from the lowest- g one, so for $R \in [0, 1]$, $V(R) = \text{base}_S(c) + R(\bar{g}(c) - \text{base}_S(c))$ — linear in R with slope $\bar{g}(c) - \text{base}_S(c) \geq 0$, hence $\sup_{R \leq 1} V(R) = V(1) = \bar{g}(c)$, and $\sup_{R \leq 1} V(R) \leq \alpha$ holds exactly under the two stated conditions. (iii) By (ii) the crossing set is an intersection of two sublevel sets of functions nondecreasing in c (i), and a sublevel set of a nondecreasing function is downward closed; the intersection of two prefixes is a prefix. Inside the sub-unit region neither condition of (ii) is binding, so τ may wobble there (and does: 323 of 792 adjacent-cap comparisons on the coarse grid); on the fine grid the crossing set has 0 holes and 0 overhangs in all 72 cells, and on the coarse grid the 323 violations decompose into 253 strictly sub-unit wobbles plus 70 upward jumps at the prefix edge, with zero holes (prefix_prop_r508_result.json). $\square$

E CROSS-CARRIER EVIDENCE MATRIX AND UNIFIED MECHANISM

Table 4 collects every in-distribution TEST readout of the paper in one place: the four model-carrier pairs (dependence/resolvable structure decreasing from left to right), all stoppers, all certified levels, and one descriptive $\alpha=0.01$ CAL-side row for OMR (where the certified family of Section 5 reports only down to $\alpha=0.02$). Validity was certified on CAL (“cert”); TEST is a single descriptive readout under the Bonferroni radius (0.0286 OMR shard 0; 0.0292 shard 1; 0.0310

OpenR1; 0.0307 RLVE). Artifacts: `fit_cal_test_r469`, `shard1_robust_r471` (R1/R2), `openr1_m2_pilot_r473`, `rlve_n8_r474`.

Table 4: Evidence matrix: carrier (dependence level) $\times$ baseline $\times$ endpoint. “null” = no certifiable budget at that level; “—” = not applicable / not reported. FULL rows are the zero-saving reference. Seeds 20260815 throughout.

Carrier	rule	α	cert	flip	mean k	saving
OMR s0 ($N=32$)	FIXED-HOEF	.10	0.0970	0.0645	7.0	78.1%
	FIXED-EB	.10	0.0958	0.0794	5.0	84.4%
	BAYES-H	.10	0.0495	0.0370	5.1	84.0%
	FIXED-HOEF	.05	0.0471	0.0156	27.0	15.6%
	FIXED-EB	.05	0.0461	0.0324	17.0	46.9%
	BAYES-H	.05	0.0350	0.0249	6.1	80.9%
	FIXED-HOEF	.02	null	—	—	—
	FIXED-EB	.02	0.0171	0.0083	31.0	3.1%
	BAYES-H	.02	0.0163	0.0091	8.5	73.5%
	BAYES-H	.01	0.0119	—	—	—
FULL-32		—	—	0	32.0	0
OMR s1 ($N=32$)	FIXED-HOEF	.05	0.0488	0.0171	25.0	21.9%
	FIXED-EB	.05	0.0486	0.0357	15.0	53.1%
	BAYES-H	.05	0.0340	0.0246	6.2	80.6%
	BAYES-H [†]	.05	0.0341	0.0247	6.2	80.6%
	BAYES-H	.02	0.0158	0.0088	8.6	73.1%
	BAYES-H [†]	.02	0.0159	0.0089	8.6	73.2%
OpenR1 ($M=2$)	FIXED-1	—	0.0707	0.0592	1.0	50.0%
	BAYES-H	.10	0.0707	0.0592	1.0	50.0%
	BAYES-H	.05	0.0383	0.0294	1.4	32.1%
	BAYES-H	.02	0.0047	0	2.0	0
RLVE ($N=8$)	FIXED-HOEF	.10	0.0942	0.0613	5.0	37.5%
	FIXED-EB	.10	0.0783	0.0613	5.0	37.5%
	BAYES-H	.10	0.0469	0.0314	3.4	57.6%
	FIXED-EB/HOEF	.05–.01	null	—	—	—
	BAYES-H	.05	0.0184	0.0103	3.9	50.9%
	BAYES-H	.02	0.0121	0.0046	4.1	48.2%
	BAYES-H	.01	0.0081	0.0019	4.4	45.4%

[†]Cross-shard prior transfer (prior fit on shard 0 FIT, TV=0.042 to the shard 1 prior); all numbers move ≤ 0.4 points against within-shard. WINDOW3 (no certificate knob) is omitted from the matrix: it saves most (86.2%) but is invalid at $\alpha \leq 0.05$ (flip 0.0577; Table 1).

Unified mechanism. One quantity — the prior’s g -profile — organizes every panel of the paper. (i) *In-distribution*, the saving tracks the resolvable mixture structure: mass on extreme- K problems stops at $k \approx 5$ (OMR, 81% saved at $\alpha=0.05$); with only $N=8$ draws and 74–87% of prior mass on zero- g atoms, three stoppable budgets remain and the saving interpolates to 45–58% (RLVE); at $M=2$ the single stoppable budget yields 32–50% and the certificate degenerates to never-stop exactly when $\max_K g(K) \leq \alpha$ (OpenR1 at $\alpha=0.02$: saving 0, flip 0 — the certificate never over-stops). (ii) *Under prior shift*, the same profile sets the critical radius: τ^* is increasing in the zero- g atom mass across carriers (Spearman 1.0 on carrier means, 16 cells; het-excess variance is not the driver: RLVE 0.0999 < OMR 0.129 yet tolerates more), with the within-OMR negative trend (−0.95/−0.89) a staircase artifact (decomposition audit, `spearman_decomposition_r486.result.json`). (iii) *Under ordered drift*, adaptivity is itself the robustness channel: BAYES-H re-reads the accumulating evidence each draw, so it is the most robust stopper under the smooth E2/E3 mechanisms (valid to $\delta=0.20$ under E2 at $\alpha=0.10$ where FIXED- k breaks at 0.15), and a margin γ on the certificate buys validity at a quantified saving cost ($\gamma=0.03$ covers E2 $\delta \leq 0.20$: flip 0.0491, saving 71.8% vs 79.7% unmarginated; $\gamma=0.025$ covers E3 $\delta \leq 0.15$: flip 0.0495, saving 77.2% vs 81.0%; `margin_repair_r469.result.json`). (iv) *The failure boundary is sharp on every axis*: premise-breaking E1 breaks all certificates together by $\delta=0.10$ –0.15; on shard 1 at $\alpha=0.01$ the observed cross-shard mismatch 0.042 exceeds $\tau^*(0.01)=0.024$ and TV alone does not certify the transfer; the coarsest fit $m=125$ sits outside its

ball (TV 0.175) and genuinely breaks at $\alpha \leq 0.02$ (Appendix D(e)). Where the paper claims a win, the margin is large and certified; where it does not, the limiting quantity is named and measured.

F REPRODUCIBILITY

All numbers trace to JSON artifacts listed in REPRO_README.md; carrier SHA-256 pinned; deterministic seeds; CPU-only; total compute under 25 minutes.